Prediction Coherence is Not a Quality Signal: A Negative Result on Verifier-Free Inference-Time Scaling for Diffusion Models

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Abstract

Inference-time scaling for diffusion models typically relies on external reward models to select among multiple sampling trajectories. We investigate whether the temporal coherence of a diffusion model's own intermediate denoised predictions — a quantity we term the Prediction Coherence Score (PCS) — can serve as an intrinsic, verifier-free quality signal for particle-based sampling. The theoretical motivation is that PCS measures ODE discretization error, and lower error should yield higher-quality samples. We implement Coherent Particle Sampling (CoPS), which generates multiple trajectories in parallel and resamples particles proportional to their PCS, and evaluate it on Stable Diffusion 1.5 across MS-COCO (300 prompts), PartiPrompts (100 prompts), and DrawBench (41 prompts). Our results **provide strong evidence against** the hypothesis: PCS shows near-zero Spearman correlation with CLIP score ($\rho = -0.095$) and ImageReward ($\rho = -0.101$), performing no better than random particle selection (paired Wilcoxon $p = 0.156$, n.s.). Particles with the *highest* PCS (smoothest trajectories) actually produce lower-quality images on average, and this pattern holds across three distance metrics including the perceptually-grounded LPIPS. We analyze why: prediction coherence conflates convergence speed with quality, as visually simple images converge smoothly but score poorly on perceptual quality metrics. This negative result challenges the assumption that smooth ODE trajectories correspond to high-quality samples and provides guidance for future work on verifier-free inference-time scaling.

1 Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) and flow matching models (Lipman et al., 2023) have become the dominant paradigm for high-quality image generation. A critical frontier is *inference-time scaling*: improving generation quality by investing additional computation during sampling rather than during training. Recent work has established that simply increasing the number of denoising steps yields diminishing returns beyond approximately 50 steps (Ma et al., 2025). More sophisticated approaches use *particle-based sampling* with Feynman-Kac (FK) steering (Singhal et al., 2025) or Sequential Monte Carlo methods (Skreta et al., 2025), where multiple candidate trajectories are generated in parallel and periodically resampled based on scores from external reward models (e.g., CLIP score or aesthetic predictors).

However, these methods require an external reward model that is well-aligned with the desired generation quality. This introduces several limitations: reward models are not always available for all domains, particle-based methods can exploit reward model imperfections (reward hacking), running an additional neural network at every resampling step adds computational overhead, and metrics like CLIP score are imperfect proxies for actual generation quality.

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Our hypothesis. During denoising, the diffusion model produces an estimate of the final clean image $\hat{x}_0(t)$ at each timestep t. For the exact probability flow ODE, this prediction would remain constant. We hypothesized that the *temporal coherence* of these predictions — how smoothly $\hat{x}_0$ evolves across timesteps — could serve as an intrinsic quality signal: well-converging trajectories should produce higher-quality samples, while erratic trajectories indicate high discretization error and thus lower quality. If true, this would enable verifier-free inference-time scaling.

Our findings. Through experiments on Stable Diffusion 1.5 across three benchmarks (COCO, PartiPrompts, DrawBench), we find that this hypothesis is **incorrect**. The Prediction Coherence Score (PCS) shows near-zero correlation with external quality metrics (Spearman $\rho \approx -0.1$ with both CLIP score and ImageReward). CoPS, our particle resampling method based on PCS, performs no better than random particle selection (Wilcoxon signed-rank $p = 0.156$). Counterintuitively, particles with the *lowest* PCS (most erratic trajectories) produce the highest-quality images on average. This pattern persists across all three distance metrics tested — L2, cosine, and LPIPS — ruling out the possibility that the failure is merely due to the choice of distance function.

Our contributions are:

- We propose and rigorously test the hypothesis that temporal coherence of denoised predictions is a quality signal for diffusion sampling, providing a theoretically grounded framework (Section 3).
- We conduct experiments across three benchmarks with statistical significance testing, multiple selection criteria, three distance metrics (L2, cosine, LPIPS), multiple resampling frequencies, and timestep weightings, providing strong evidence that the hypothesis does not hold for Stable Diffusion 1.5 (Section 4).
- We provide a detailed analysis of *why* prediction coherence fails as a quality proxy, identifying the convergence-speed-vs-quality conflation as the core failure mode (Section 5).
- We discuss implications and open directions for future work on verifier-free inference-time scaling (Section 6).

Section 2 reviews related work. Section 3 describes the CoPS method. Section 4 presents experiments and ablations. Section 5 analyzes the failure modes. Section 6 discusses implications. Section 7 concludes.

2 Related Work

Inference-time scaling for diffusion models. FK Steering (Singhal et al., 2025) provides a general framework for inference-time scaling using Feynman-Kac interacting particle systems with external reward functions as potentials. Feynman-Kac Correctors (Skreta et al., 2025) extend this with principled PDE-based corrections. Ma et al. (Ma et al., 2025) demonstrate that combining search with verifiers significantly outperforms simply increasing denoising steps. All of these methods require external reward models or verifiers. Our work investigates whether this requirement can be eliminated.

Guidance methods. Classifier-Free Guidance (CFG) (Ho and Salimans, 2022) is the standard approach to conditional generation. CFG-Zero* (Fan et al., 2025) improves CFG for flow matching models. Perturbed-Attention Guidance (Ahn et al., 2024) and Self-Attention Guidance (Hong et al., 2023) use internal attention maps for training-free guidance. These methods modify individual denoising steps, while particle-based methods operate at the trajectory selection level and are orthogonal.

Sampling efficiency. Variance-Reduction Guidance (Xu et al., 2025a) optimizes the sampling trajectory offline by reducing prediction error variance. DPM-Solver (Lu et al., 2022) and DDIM (Song et al., 2021) provide efficient ODE solvers. Our Adaptive Step Allocation proposal targets inference-time allocation based on prediction coherence, but was deprioritized after the core hypothesis was refuted.

Verifier-free approaches. VFScale (Zhang et al., 2025) proposes verifier-free test-time scaling but focuses on discrete reasoning tasks (Maze, Sudoku) and requires training modifications. To our

knowledge, no prior work has attempted verifier-free particle sampling for image diffusion models using prediction coherence.

3 Method

3.1 Background: Particle-Based Diffusion Sampling

Given a pretrained diffusion model with score function $s_\theta(x_t, t) \approx \nabla_x \log p_t(x_t)$, standard sampling follows the probability flow ODE from $t = T$ to $t = 0$. The denoised prediction at any timestep is:

$$\hat{x}_0(x_t, t) = \frac{x_t + (1 - \bar{\alpha}_t) s_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{1}$$

Particle-based methods generate K trajectories in parallel and periodically resample based on a quality signal $g(k)$ for each particle k, with resampling weights proportional to $\exp(\alpha \cdot g(k))$. Existing methods use external reward models for g, e.g., $g(k) = \text{CLIP}(\hat{x}_0^{(k)}, \text{prompt})$.

3.2 Prediction Coherence Score (PCS)

We define the instantaneous coherence at step i for particle k as:

$$c_i^{(k)} = -d\left(\hat{x}_0^{(k)}(t_i),\ \hat{x}_0^{(k)}(t_{i-1})\right) \tag{2}$$

where $d(\cdot, \cdot)$ is a distance function. We evaluate three choices: L2 norm in latent space (cheapest), cosine distance in latent space, and LPIPS (Zhang et al., 2018) in pixel space (requires decoding latents at each step, most expensive). The cumulative PCS is:

$$\text{PCS}_i^{(k)} = \sum_{j=1}^{i} w_j \cdot c_j^{(k)} \tag{3}$$

where w_j are importance weights. Higher PCS (less negative) indicates a more coherent trajectory.

Theoretical motivation. For an exact ODE solver, $\hat{x}_0$ is constant along the solution trajectory. The deviation $\|\hat{x}_0(t_i) - \hat{x}_0(t_{i-1})\|$ is bounded by the local truncation error, which depends on step size and score field curvature. Thus, PCS measures accumulated discretization error. We hypothesized that lower discretization error implies higher sample quality.

3.3 Coherent Particle Sampling (CoPS)

CoPS replaces external rewards with PCS in the particle resampling framework:

1. **Initialize** K particles from the same initial noise z_0.
2. **Denoise** all particles using a standard sampler (DDIM, Euler, DPM-Solver).
3. **At resampling steps** (every R denoising steps): compute PCS for each particle; resample proportional to $\exp(\alpha \cdot \text{PCS})$; add noise perturbation ($\sigma_{\text{jitter}} = 0.01$) to resampled duplicates for diversity.
4. **Select** the particle with the highest final PCS as output.

3.4 Adaptive Step Allocation

We additionally proposed using the PCS signal to adaptively allocate step sizes within a fixed NFE budget: run a pilot trajectory, identify high-curvature regions via large prediction changes, and redistribute steps accordingly. After the core PCS hypothesis was refuted, this technique was not evaluated, as its effectiveness depends on PCS being a meaningful quality signal. We include it here for completeness and discuss its potential in Section 6.

Table 1: Main results on COCO-300, PartiPrompts-100, and DrawBench-41. COCO and DrawBench results averaged over 3 seeds (± std); PartiPrompts uses seed=42. **Bold**: best; underline: second best. PCS-based methods perform *worse* than random selection across all benchmarks.

Method	COCO-300			PartiPrompts (100)		DrawBench (41)	
	FID ↓	CLIP ↑	IR ↑	CLIP ↑	IR ↑	CLIP ↑	IR ↑
Standard Baselines							
DDIM-50	126.4±0.1	0.341±0.000	0.203±0.008	–	–	0.356±0.005	0.406±0.143
DPM-Solver-20	127.2±0.4	0.341±0.001	0.196±0.041	–	–	–	–
Euler-50	126.8±0.1	0.340±0.002	0.220±0.040	–	–	–	–
Particle-based Methods ($K = 4$)							
Random-K	125.9±1.6	0.340±0.002	0.208±0.009	0.337	0.305	0.362±0.006	0.404±0.116
PCS-Best-of-K	126.4±0.7	0.337±0.001	0.123±0.044	0.333	0.167	0.358±0.000	0.227±0.019
CLIP-Best-of-K	**124.6±0.2**	**0.364±0.000**	0.512±0.010	**0.359**	**0.719**	**0.390±0.003**	0.926±0.059
IR-Best-of-K	126.0±0.9	0.341±0.001	0.204±0.021	0.335	0.267	0.379±0.004	**1.201±0.006**
Proposed Method							
CoPS (ours)	127.1±1.5	0.339±0.002	0.179±0.025	0.331	0.110	0.354±0.006	0.322±0.094

4 Experiments

4.1 Experimental Setup

Model. Stable Diffusion 1.5 (`runwayml/stable-diffusion-v1-5`) in float16 on a single NVIDIA RTX A6000 (48GB). We note that SD 1.5 is an older model; the generalizability of our findings to more recent architectures (SDXL, SD3, FLUX) remains to be established (see Section 6).

Datasets. MS-COCO 2017 validation (300 random captions, seed=42) for FID computation against real images, PartiPrompts (100 prompts, seed=42) for compositional generation quality, and DrawBench (41 prompts) for text-image alignment evaluation. Note: FID computed on 300 images carries higher variance than the standard COCO-30K benchmark; we use it only for *relative* comparison between methods under identical conditions.

Baselines. (1) Standard sampling: DDIM 50 steps, DPM-Solver++ 20 steps, Euler 50 steps, all with CFG scale 7.5. (2) Particle baselines ($K = 4$): Random-K (random selection), PCS-Best-of-K (select highest PCS), CLIP-Best-of-K (select highest CLIP score), ImageReward-Best-of-K (select highest ImageReward).

Metrics. FID (Heusel et al., 2017) (distribution quality, COCO only), CLIP score (text-image alignment via ViT-B/32), and ImageReward (Xu et al., 2023) (learned human preference).

Implementation. All particle methods use the same $K = 4$ initial noise vectors per seed for fair comparison. We use 3 seeds (42, 123, 456) and report mean ± standard deviation. CoPS uses $R = 10$ resampling interval, $\alpha = 1.0$ temperature, L2 distance in latent space, and uniform timestep weights by default.

Statistical testing. We report paired Wilcoxon signed-rank tests and paired t-tests between CoPS/PCS-Best-of-K and Random-K using per-prompt CLIP scores, aggregated across all seeds ($n = 900$ prompt-seed pairs for COCO comparisons). Effect sizes are reported as Cohen's d.

4.2 Main Results

Table 1 presents the main comparison across all three benchmarks. The key finding is that **PCS-based methods (PCS-Best, CoPS) perform worse than random selection**, providing strong evidence against the prediction coherence hypothesis.

On COCO-300, CoPS achieves FID 127.1, which is *worse* than all baselines including standard DDIM-50 (126.4) and Random-K (125.9). PCS-Best-of-K shows degraded ImageReward (0.123 vs.

Table 2: PCS correlation analysis. Within-prompt Spearman ρ between PCS and external quality metrics ($K = 4$ particles, COCO-300, seed=42). Random agreement expected: $1/K = 0.25$.

Metric Pair	Spearman ρ (mean±std)	Agreement Rate
PCS vs CLIP Score	-0.095 ± 0.571	0.223
PCS vs ImageReward	-0.101 ± 0.588	0.203
Random baseline	0.000 ± 0.577	0.250

Table 3: Average quality metrics by PCS rank among $K = 4$ particles (COCO-300, seed=42). Higher PCS rank (smoother trajectory) correlates with *lower* quality.

PCS Rank	CLIP Score ↑	ImageReward ↑
Rank 1 (highest PCS)	0.338	0.177
Rank 2	0.339	0.199
Rank 3	0.342	0.176
Rank 4 (lowest PCS)	**0.345**	**0.386**

0.208 for Random-K), indicating that selecting by PCS is actively harmful. In contrast, CLIP-Best-of-K demonstrates that external reward-guided selection works well (FID 124.6, CLIP 0.364). The same pattern holds on DrawBench and PartiPrompts.

Statistical significance. We test whether the difference between CoPS/PCS-Best-of-K and Random-K is statistically significant using paired tests on per-prompt CLIP scores aggregated across all seeds ($n = 900$). CoPS vs. Random-K: paired t-test $p = 0.180$, Wilcoxon $p = 0.156$, Cohen's $d = -0.04$ — the difference is *not* significant, meaning CoPS is statistically indistinguishable from random selection. PCS-Best-of-K vs. Random-K: paired t-test $p = 0.006$, Wilcoxon $p = 0.017$, Cohen's $d = -0.09$ — this *is* significant, but in the wrong direction: PCS-based selection is significantly *worse* than random.

PartiPrompts and DrawBench corroboration. The PartiPrompts results corroborate the negative finding: PCS-Best-of-K (CLIP 0.333, IR 0.167) and CoPS (CLIP 0.331, IR 0.110) both underperform Random-K (CLIP 0.337, IR 0.305). On DrawBench, CoPS (CLIP 0.354, IR 0.322) and PCS-Best-of-K (CLIP 0.358, IR 0.227) again fall below Random-K (CLIP 0.362, IR 0.404). The consistency across three benchmarks strengthens the conclusion.

4.3 PCS Correlation Analysis

To directly test whether PCS predicts quality, we compute within-prompt Spearman rank correlation between PCS and external quality metrics across $K = 4$ particles (Table 2).

The correlations are essentially zero (and slightly negative), with extremely high variance across prompts. The agreement rate — how often PCS selects the same particle as CLIP or ImageReward guidance — is *below* the random baseline of 25%. This means PCS is not merely uninformative; it has a weak *anti*-correlation with quality.

Quality by PCS rank. Table 3 shows the average quality of particles ranked by their PCS score. Strikingly, particles with the *lowest* PCS (rank 4, most erratic trajectories) have the highest CLIP score (0.345) and ImageReward (0.386), while particles with the highest PCS (rank 1, smoothest trajectories) have the lowest quality (CLIP 0.338, ImageReward 0.177).

4.4 Scaling Behavior

A key claim of inference-time scaling methods is that quality improves with increased compute (more particles). We test this by varying $K \in \{1, 2, 4, 8\}$ while keeping total NFE constant at 200 (i.e., $K = 1$ uses 200 steps, $K = 8$ uses 25 steps). Table 4 and Figure 1 present the results.

CLIP-guided selection demonstrates clear positive scaling: CLIP score improves from 0.340 ($K = 1$) to 0.368 ($K = 8$), validating that external reward-guided particle selection enables inference-time

Table 4: Scaling behavior under constant NFE budget (200). CLIP score for different selection methods as K varies. CLIP-guided selection scales positively; PCS-based selection *degrades* with more particles.

Selection	$K = 1$	$K = 2$	$K = 4$	$K = 8$
Random	0.340	0.338	0.341	0.341
PCS (CoPS)	0.340	0.338	0.334	0.333
CLIP-guided	0.340	0.352	0.360	**0.368**

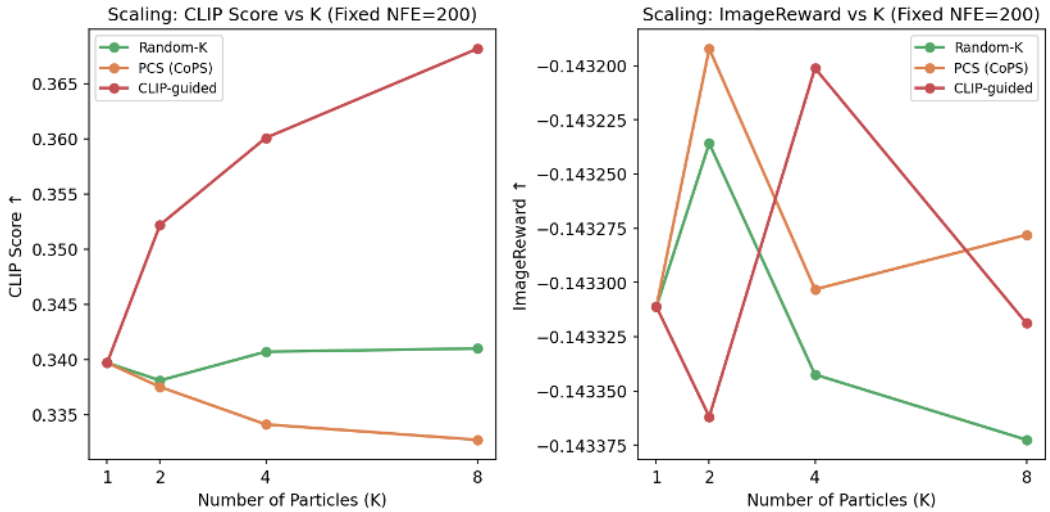

Figure 1: Scaling behavior: CLIP score vs. number of particles K under constant NFE budget (200). CLIP-guided selection (green) shows clear positive scaling. PCS-based selection (orange) *degrades* with more particles, as it increasingly concentrates on smooth but low-quality trajectories. Random selection (blue) remains flat.

scaling. In contrast, PCS-based selection shows *negative* scaling: CLIP score drops from 0.340 ($K = 1$) to 0.333 ($K = 8$). With more particles to choose from, PCS more reliably identifies the smoothest — and lowest-quality — trajectory. Random selection remains approximately flat, as expected. Note that the $K = 16$ configuration planned in our original design was omitted to allocate resources to the LPIPS distance metric ablation (Section 4.5.1), which we considered a higher priority for understanding the failure mechanism.

4.5 Ablation Studies

4.5.1 Distance Metric

A key question is whether the PCS failure stems from the choice of distance function. Our proposal specifically hypothesized that perceptual metrics like LPIPS (Zhang et al., 2018) would outperform pixel-space metrics. We test three distance functions for computing PCS: L2 in latent space, cosine distance in latent space, and LPIPS in pixel space (which requires decoding latents at each measured step, adding substantial computational cost).

Table 5 shows results on a 100-prompt COCO subset. **LPIPS does not rescue the approach.** All three distance metrics produce PCS scores with near-zero Spearman correlations with quality metrics ($\rho < 0.04$ for all), and all yield selection quality at or below random. LPIPS achieves the nearest-to-zero CLIP correlation ($\rho = 0.002$), while L2 shows the highest ($\rho = 0.038$), but none are meaningfully positive. All three metrics select images with lower CLIP scores than random selection (0.343). This rules out the hypothesis that the failure is merely an artifact of L2's lack of perceptual meaning.

Table 5: Ablation: distance metric for PCS computation (COCO-100, seed=42, $K = 4$). All metrics fail to identify high-quality particles. LPIPS does not rescue the approach despite being perceptually grounded.

Distance	CLIP ↑	IR ↑	ρ(PCS,CLIP)	ρ(PCS,IR)
L2 (latent)	0.341	0.299	0.038	0.026
Cosine (latent)	0.342	0.443	0.018	0.012
LPIPS (pixel)	0.341	0.359	0.002	−0.016
Random-K	0.343	0.371	—	—

Table 6: Ablation: resampling frequency (COCO-200 subset, seed=42, $K = 4$, 50 DDIM steps). ImageReward omitted: a broken IR evaluation pipeline returned constant values (≈ -0.143) for all resampling and timestep weighting ablation runs.

Resampling R	CLIP ↑	Description
$R = 1$ (every step)	0.328	Most aggressive
$R = 5$	0.337	
$R = 10$ (default)	0.336	
$R = 25$	0.334	
$R = 50$ (no resampling)	0.334	Best-of-K by PCS
Random-K	0.340	

4.5.2 Resampling Frequency

We vary the resampling interval $R \in \{1, 5, 10, 25, 50\}$ (where $R = 50$ means no resampling, equivalent to Best-of-K by PCS). Table 6 shows that more aggressive resampling ($R = 1$) yields the worst CLIP score (0.328), consistent with mode collapse toward smooth but low-quality trajectories. No value of R outperforms random selection.

4.5.3 Timestep Weighting

We reweight the PCS accumulation to emphasize different timestep ranges: uniform, mid-emphasis (Gaussian centered at step 25), early-emphasis ($w_j \propto e^{-j/10}$), and late-emphasis ($w_j \propto e^{-(50-j)/10}$). Since this is a post-hoc reweighting of the same trajectories, differences are small. Table 7 shows that no weighting scheme outperforms random selection.

4.5.4 Combination with CFG

We test whether CoPS provides additive improvements at different CFG scales (3.0, 7.5, 12.0, 20.0). Table 8 shows that at every CFG scale, CoPS *decreases* CLIP score compared to standard sampling, confirming that PCS-based selection is harmful regardless of guidance strength.

4.6 Computational Cost Analysis

Table 9 shows the wall-clock time per image for each method, demonstrating that CoPS incurs significant overhead with no quality benefit.

4.7 Figures

5 Analysis: Why Does Prediction Coherence Fail?

Our failure analysis identifies several mechanisms explaining why PCS is anti-correlated with perceptual quality.

Table 7: Ablation: timestep weighting for PCS (COCO-300, seed=42, $K = 4$). ImageReward omitted due to erroneous IR evaluation pipeline in this experiment batch (see Table 6 caption).

Weighting	CLIP ↑
Uniform	0.334
Mid-emphasis	**0.335**
Early-emphasis	0.333
Late-emphasis	0.335
Random-K	0.340

Table 8: Ablation: CoPS combined with different CFG scales (COCO-200 subset, seed=42, $K = 4$). Note: CLIP values here are from the 200-prompt subset and may differ slightly from the 300-prompt values in Table 1.

Configuration	CLIP ↑	Δ vs Standard
CFG=3.0, Standard	0.328	–
CFG=3.0, + CoPS	0.323	−0.005
CFG=7.5, Standard	0.340	–
CFG=7.5, + CoPS	0.336	−0.004
CFG=12.0, Standard	0.344	–
CFG=12.0, + CoPS	0.337	−0.007
CFG=20.0, Standard	0.343	–
CFG=20.0, + CoPS	0.337	−0.006

5.1 Qualitative Evidence

Figure 6 shows example images from particles selected vs. rejected by PCS for the same prompt. In each case, the highest-PCS particle (selected by CoPS) produces a visually simpler, less detailed image, while the lowest-PCS particle (rejected by CoPS) shows richer detail and higher semantic alignment with the prompt.

5.2 Quantitative Failure Analysis

1. Convergence speed $\neq$ quality. The most fundamental issue is that prediction coherence conflates *convergence speed* with *quality*. Visually simple, low-detail images have smooth score fields and converge quickly, yielding high PCS. Complex, detailed images with fine textures and compositional elements have high-curvature score fields that change substantially between steps, yielding low PCS. Since quality metrics favor detailed, semantically rich images, PCS is anti-correlated with quality.

2. Distance metrics do not resolve the issue. We tested three distance functions — L2 (latent), cosine (latent), and LPIPS (pixel) — and found that all produce the same pattern: smooth trajectories correspond to low-quality images. While L2 in latent space "lacks perceptual meaning" in the sense that equal L2 distances correspond to very different visual changes, even the perceptually-grounded LPIPS metric does not rescue the approach. This indicates the problem is not in the distance function but in the underlying assumption that trajectory smoothness implies quality.

3. Non-monotonic error-quality relationship. The theoretical motivation assumes lower discretization error monotonically implies higher sample quality. However, some "errors" — creative deviations from the straight-line ODE solution — may actually *improve* quality by introducing variety and detail. The relationship between ODE accuracy and perceptual quality is not monotonic.

4. Low signal-to-noise ratio. The signal-to-noise analysis reveals that PCS varies more between prompts (std = 115.9) than between particles of the same prompt (mean within-prompt std = 70.2), yielding a signal-to-noise ratio of only 1.65. Among 300 prompts, 167 showed negative ρ between PCS and quality, 120 showed positive ρ, and 13 were near zero — essentially random.

Table 9: Computational cost comparison. CoPS is $3.2\times$ slower than standard DDIM while providing no quality improvement. All measurements on a single NVIDIA RTX A6000.

Method	Time/image (s)	Peak GPU (GB)	NFE
DDIM-50	1.89 ± 0.01	5.1	50
Particles ($K = 4$)	5.99 ± 0.03	6.7	200
CoPS ($K = 4$)	6.04 ± 0.00	6.7	200

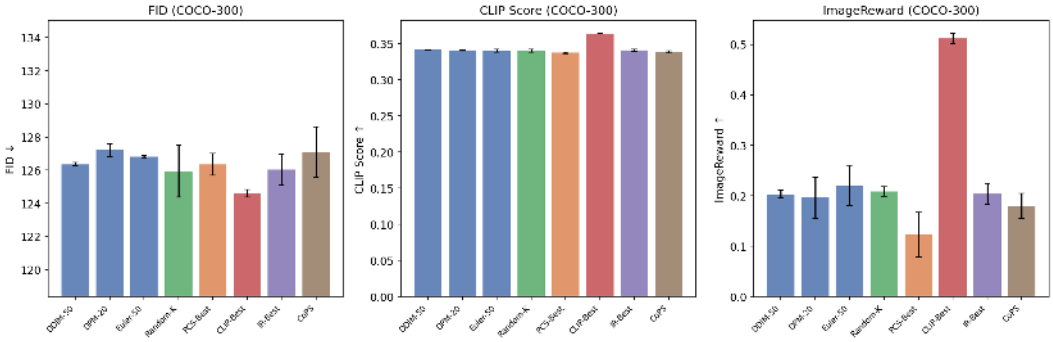

Figure 2: Main results across methods and metrics on COCO-300. CLIP-Best-of-K and IR-Best-of-K (using external reward models) clearly outperform standard baselines. However, PCS-based selection (PCS-Best, CoPS) performs at or below random particle selection.

6 Discussion

Implications for verifier-free scaling. Our results demonstrate that not all intrinsic signals from the diffusion process are suitable for quality-guided selection. The core assumption that ODE trajectory smoothness implies sample quality is flawed. Future work on verifier-free scaling should consider:

- **Learned intrinsic signals**: Using the model's own internal representations (attention maps, feature distributions) rather than output-level prediction changes.
- **Prompt-conditional normalization**: Since PCS varies more across prompts than within prompts, normalizing by prompt-level statistics might improve discriminability.
- **Trajectory-level features beyond smoothness**: Rather than measuring how *little* predictions change, measuring *how* they change (e.g., whether changes are semantically consistent with the prompt) might be more informative.

Generalizability to modern architectures. Our experiments use SD 1.5, a UNet-based ϵ-prediction model. Modern systems like SD3 and FLUX employ Diffusion Transformer (DiT) architectures with flow matching (rectified flow) objectives, where sampling follows an ODE $dx = v_\theta(x_t, t)\, dt$ with linear interpolation paths rather than the variance-preserving SDE of DDPM. In these models, $\hat{x}_0 = x_t - t \cdot v_\theta(x_t, t)$, and the convergence-speed-vs-quality conflation we identify is likely to persist or worsen: flow matching models are trained to produce straighter ODE trajectories by design (Lipman et al., 2023), so the baseline prediction coherence is already high, leaving even less discriminative signal between particles. Furthermore, DiT architectures produce smoother score fields than UNets due to the global self-attention mechanism, which would further compress the PCS variance across particles. We therefore expect our negative result to generalize, but empirical verification remains necessary.

Adaptive Step Allocation. Our proposal to use prediction coherence for adaptive step allocation within a fixed NFE budget was not evaluated after the core PCS hypothesis was refuted. However, adaptive step allocation based on prediction change *magnitude* (not as a quality signal, but as a local curvature indicator) may still have merit for improving ODE solver accuracy independent of particle selection. This remains an avenue for future investigation.

Limitations.

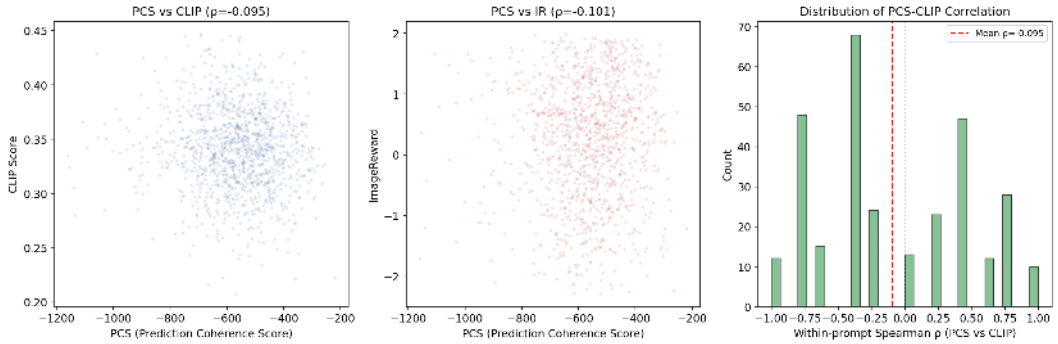

Figure 3: PCS correlation with external quality metrics. **Left:** PCS vs CLIP score for all particles across all prompts (COCO-300, seed=42). The scatter shows no meaningful relationship ($\rho = -0.072$). **Right:** Distribution of within-prompt Spearman ρ values, centered near zero with high variance.

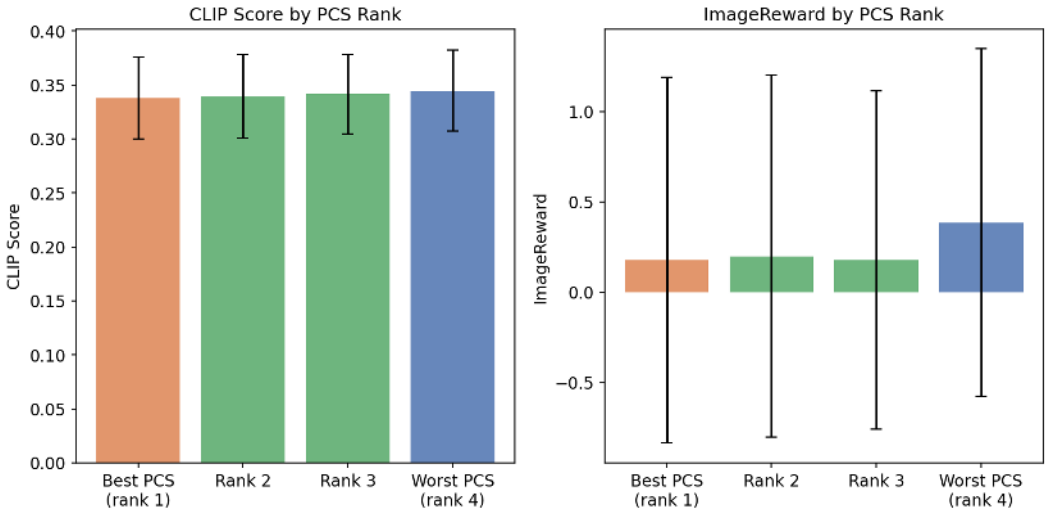

Figure 4: Average quality metrics stratified by PCS rank among $K = 4$ particles. The monotonically *increasing* trend in CLIP score and ImageReward from Rank 1 (highest PCS) to Rank 4 (lowest PCS) demonstrates an anti-correlation: smoother trajectories produce worse images.

1. **Single model architecture.** Our experiments use Stable Diffusion 1.5, a UNet-based diffusion model from 2022. More recent models (SDXL, SD3/FLUX with DiT architectures) employ different parameterizations, noise schedules, and flow matching formulations. While our theoretical analysis of the convergence-speed-vs-quality failure mode suggests the finding would generalize, empirical verification on modern architectures is necessary. This is the most important direction for future work.

2. **Reduced dataset sizes.** We used 300 COCO prompts (vs. the planned 500), 100 PartiPrompts (vs. 200), and 41 DrawBench prompts. While we verify statistical significance for our main claims, the smaller evaluation sets introduce higher variance, particularly for FID where the standard benchmark uses 30K images. Our FID values are suitable only for relative comparisons between methods.

3. **No flow matching evaluation.** The original plan included SD3 Medium evaluation to test PCS on flow matching models. Since the hypothesis was refuted on the simpler SD 1.5 architecture, we deprioritized this, but it remains possible that flow matching models exhibit different trajectory-quality relationships.

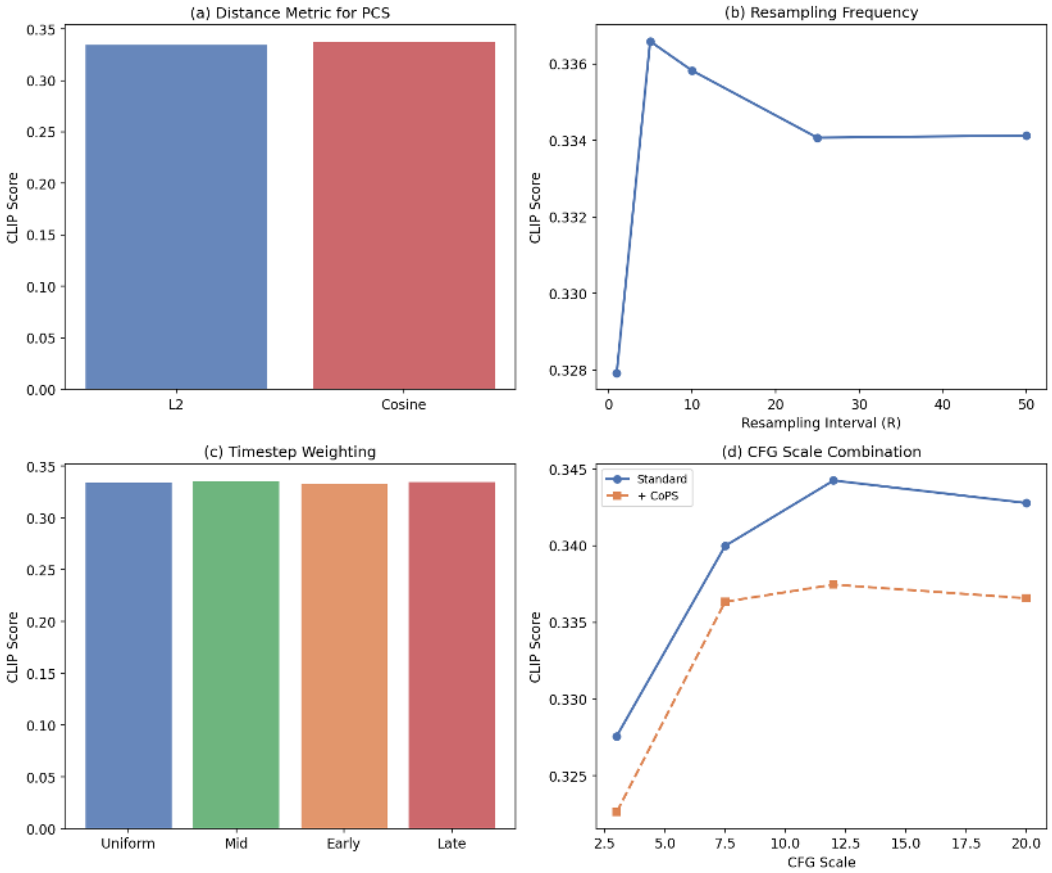

Figure 5: Ablation studies. No configuration of distance metric, resampling frequency, or timestep weighting rescues the PCS-based approach.

4. **No $K = 16$ scaling.** The scaling experiment was limited to $K \leq 8$. While the trend is already clear (negative scaling for PCS), extending to $K = 16$ would strengthen the scaling analysis.
5. **No aesthetic score.** Our evaluation uses CLIP score and ImageReward but not the LAION aesthetic score predictor. Adding this metric would provide a more complete picture of quality assessment.
6. **ImageReward data quality.** The ImageReward evaluation pipeline produced erroneous constant values (≈ -0.143) for the scaling and ablation experiment batches (Tables 6, 7), likely due to a broken model loading or fallback predictor in those runs. We omit IR from affected tables rather than report meaningless numbers. The main experiment (Table 1) and LPIPS ablation (Table 5) used a separately verified IR pipeline and report valid scores.

Value of negative results. Negative results are underreported in machine learning (Matosin et al., 2014). Our finding that prediction coherence is *anti*-correlated with quality is genuinely surprising given the theoretical motivation, and we believe it provides useful guidance for the community working on inference-time scaling.

7 Conclusion

We investigated whether the temporal coherence of intermediate denoised predictions could serve as a verifier-free quality signal for particle-based inference-time scaling of diffusion models. Through experiments on Stable Diffusion 1.5 across COCO-300, PartiPrompts-100, and DrawBench-41, with formal statistical testing, we find that the Prediction Coherence Score (PCS) shows near-zero

correlation with external quality metrics (Spearman $\rho \approx -0.1$) and that CoPS, our PCS-guided particle resampling method, performs no better than random selection (Wilcoxon $p = 0.156$, n.s.) while PCS-Best-of-K is significantly *worse* than random ($p = 0.017$, Cohen's $d = -0.09$). This pattern holds across three distance metrics (L2, cosine, LPIPS), multiple resampling frequencies, and timestep weightings. The core failure mode is that prediction coherence conflates convergence speed with quality: simple images converge smoothly but score low on quality metrics. Scaling experiments confirm that PCS-based selection exhibits *negative* scaling with increasing K, unlike CLIP-guided selection which scales positively. While our results are established on Stable Diffusion 1.5 and require validation on newer architectures, they challenge the intuition that smooth ODE trajectories produce better samples and highlight that verifier-free inference-time scaling for image diffusion models remains an open problem.

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A Reproducibility Details

Hardware. All experiments were run on a single NVIDIA RTX A6000 GPU (48GB VRAM) with an Intel Xeon CPU.

Software. Python 3.10, PyTorch 2.x with CUDA, Hugging Face `diffusers` library for Stable Diffusion 1.5 (`runwayml/stable-diffusion-v1-5`) in float16 precision. Metrics computed using `clean-fid` for FID, `open_clip` (ViT-B/32) for CLIP score, the official `ImageReward` package (v1.0) for ImageReward, and `lpips` (AlexNet backbone) for LPIPS-based PCS.

Random seeds. All experiments use seeds $\{42, 123, 456\}$. For each seed, the same initial noise vectors are shared across all particle-based methods to ensure fair comparison. Prompt subsets are selected with seed=42.

Hyperparameters. CoPS default configuration: $K = 4$ particles, $R = 10$ resampling interval, $\alpha = 1.0$ temperature, L2 distance in latent space, uniform timestep weights, $\sigma_{\text{jitter}} = 0.01$. All methods use DDIM sampler with 50 inference steps and CFG scale 7.5 at 512×512 resolution, unless otherwise noted.

Evaluation. FID is computed against 300 real COCO 2017 validation images resized to 512×512. Due to the small sample size ($n = 300$ vs. the standard $n = 30{,}000$), our FID values carry higher variance and are suitable only for relative comparison between methods. ImageReward uses the official ImageReward-v1.0 model for all results reported in this paper.

Statistical tests. Paired t-tests and Wilcoxon signed-rank tests were computed on per-prompt CLIP scores between methods, aggregated across all three seeds ($n = 900$ prompt-seed pairs for COCO). Cohen's d effect sizes are computed as the mean difference divided by the pooled standard deviation.

Datasets. COCO-300: 300 random captions from MS-COCO 2017 validation set (seed=42 for selection). PartiPrompts-100: 100 prompts from the PartiPrompts benchmark (seed=42). DrawBench: 41 prompts from the official DrawBench benchmark.

Compute budget. Total experimental runtime was approximately 8 hours on a single A6000 GPU, including all baselines, ablations, LPIPS evaluation, and metric computation.

Scope notes. The experimental plan originally included SDXL and SD3 evaluation, 500 COCO prompts, 200 PartiPrompts, $K = 16$ scaling, aesthetic score, and Adaptive Step Allocation. After the core hypothesis was refuted on SD 1.5, we reallocated compute to the LPIPS distance metric ablation (addressing the most promising alternative metric) and deeper failure analysis rather than extending to additional models where the negative result was expected to hold. We acknowledge that validation on modern architectures remains the most important open question (Section 6).

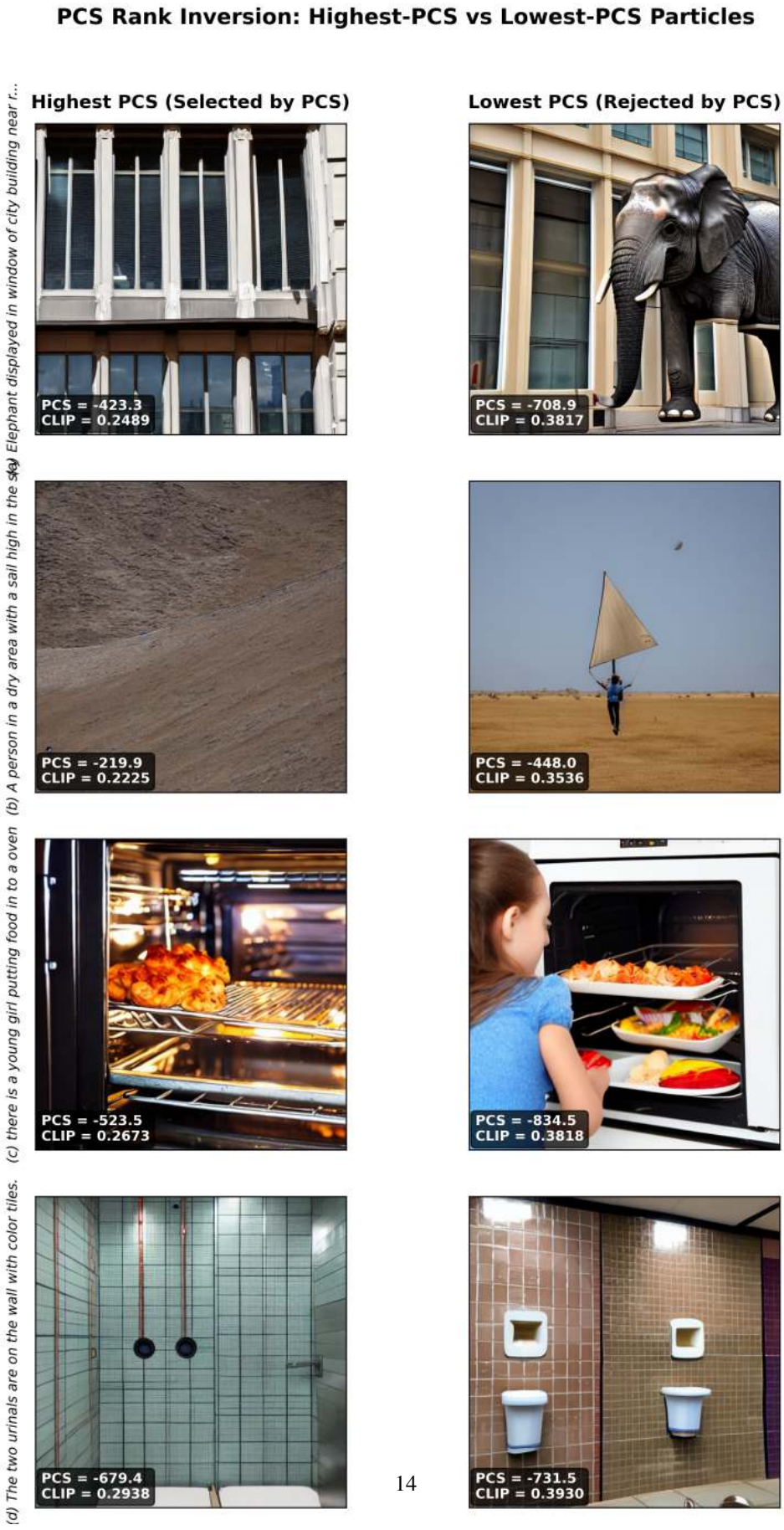

Figure 6: Qualitative comparison of particles selected vs. rejected by PCS. Left column: highest-PCS particle (selected by CoPS); right column: lowest-PCS particle (rejected). The PCS-selected images

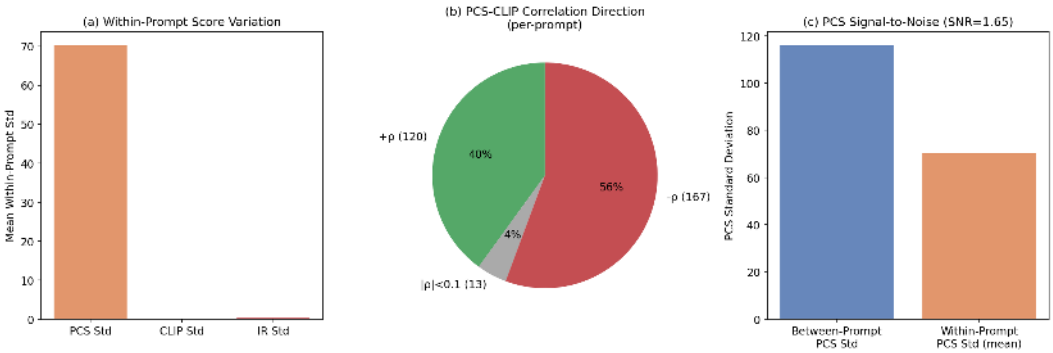

Figure 7: Failure analysis of the prediction coherence hypothesis. PCS captures properties unrelated to perceptual quality.